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Automotive Cyber Security Academy 2023

Chapter 4 – Automotive Hardware Security

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Introduction

Understanding the Risks

SW/FW manipulation

Personal damage: safety impact on the road user

Asset: FW

The device/system SHALL **ensure** that only *trusted* (**authentic**) code is loaded during the **boot process**.

Authenticity — Only from approved **origin**

CS properties

Integrity — Only with approved **content**

The device/system SHALL be **protected** from loading **manipulated** or **untrusted** (**unauthentic**) code during the **boot process**.

Spoofing

Tampering

Hardware Supported Security

- Local and remote software attacks
  Security-critical assets shielded in hardware
- Insider, offline, physical tampering attacks
  Physical tamper protection
- High performance security requirements
  Cryptographic hardware accelerators
- Costly to extend general-purpose hardware
  Cost-efficient special optimized circuits

Benefits of Hardware-based Security

	Software Only	Hardware Support
	Protects against limited attackers and basic software attacks	Protects against hardware attacks and hardens against software attacks
Reading	Software easily readable	Protection against unauthorized reading
Copying	Software can easily be copied and distributed	Secure hardware cannot be easily copied
Analyzing	Software easily analyzable using standard tools	Secure hardware implements mechanisms to impede analyzes
Root of Trust	No "Root of Trust"	"Root of Trust" which provides detection, recoverability …

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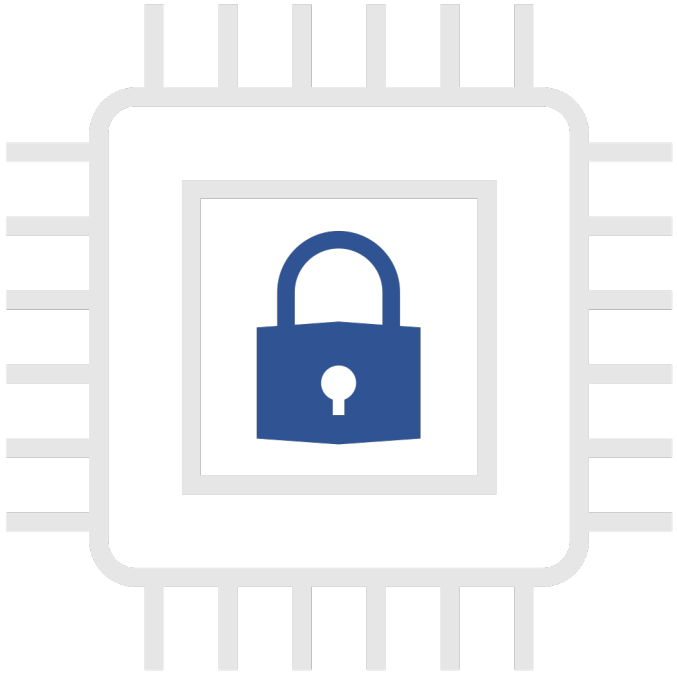
Physical Security Measures

- Tamper Evidence
  - Provide detection if a hardware component is compromised
  - Detection of unsuccessful attempts, e.g., special labels
- Tamper Resistance or Tamper Proof
  - Passively prevent an attacker from illicitly modifying or compromising hardware
  - Hardened steel, special sealing, special material against side channel attacks, very small semiconductor structures (90nm technology or less)
  - Logical measures: bus and memory encryption, obfuscation, check integrity of information on every power-up
- Tamper Response
  - Once tampering has been detected actively trigger measure to protect assets
    - Interruption of security critical operations
    - Complete deactivation
    - Deletion of internal secrets by RAM power drop or RAM overwrite
    - Requires internal autonomous power supply, e.g., long-life batteries

Summary

- Hardware-based security should …
  - … make it at least difficult for the attacker
  - … ensure that compromise of a single component does not compromise the whole system.
  - … make the cost for compromise higher than the potential rewards.



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Thank You

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